INSTALLER / ELECTRICIAN NOTE:
EC IS TO MEASURE VOLTAGE BEFORE STARTING WORK.
IF RESULT IS ANY OTHER VOLTAGE MEASURED THAN 120/240V IS OBSERVED, DO NOT PROCEED. CONTACT ENGINEER.

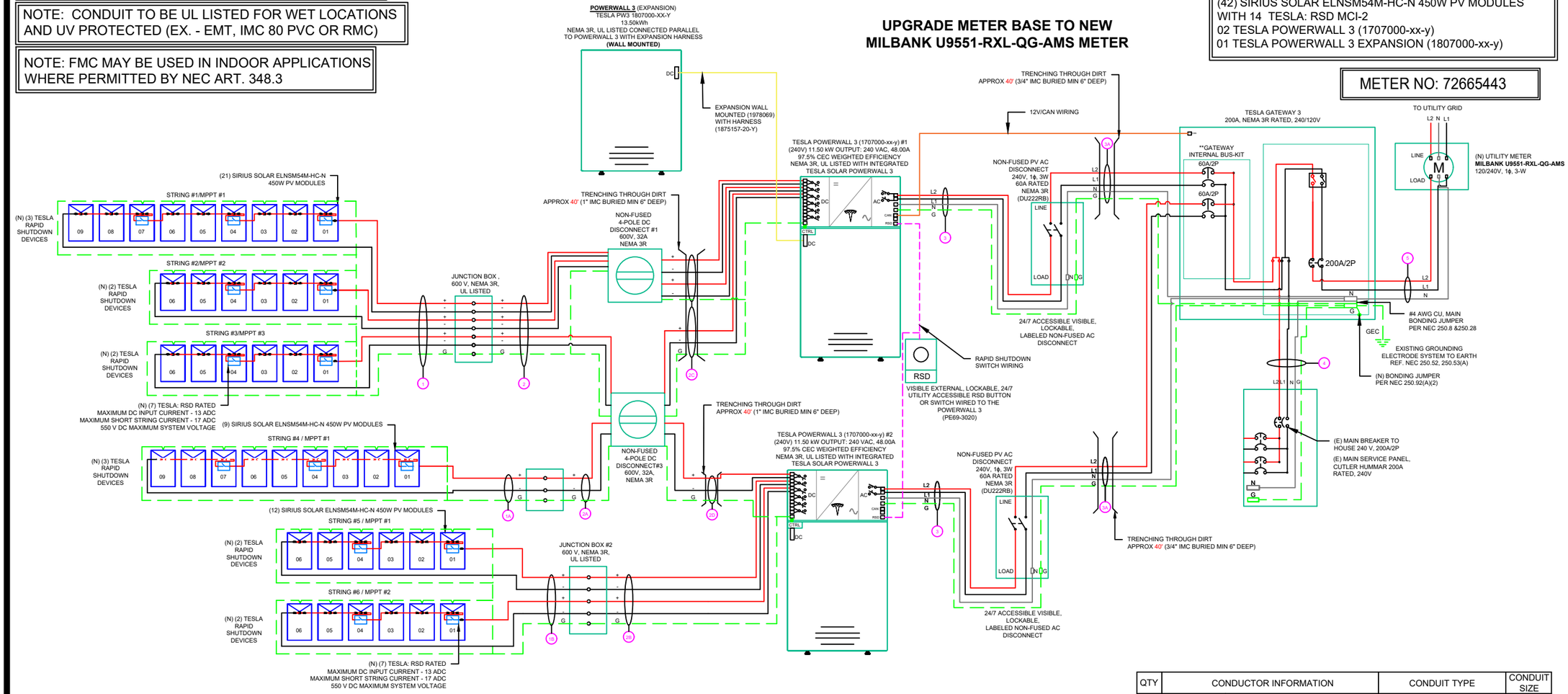
NOTE: CONDUIT TO BE UL LISTED FOR WET LOCATIONS AND UV PROTECTED (EX. - EMT, IMC 80 PVC OR RMC)

NOTE: FMC MAY BE USED IN INDOOR APPLICATIONS WHERE PERMITTED BY NEC ART. 348.3

UPGRADE METER BASE TO NEW MILBANK U9551-RXL-QG-AMS METER

DC SYSTEM SIZE: 42 x 450 = 18.900KW DC
AC SYSTEM SIZE: 02 x 11500 = 23.000KW AC
ESS AC CAPACITY: 40.5 KWH
(42) SIRIUS SOLAR ELNSM54M-HC-N 450W PV MODULES WITH 14 TESLA: RSD MCI-2
02 TESLA POWERWALL 3 (1707000-xx-y)
01 TESLA POWERWALL 3 EXPANSION (1807000-xx-y)

METER NO: 72665443



	QTY	CONDUCTOR INFORMATION		CONDUIT TYPE	CONDUIT SIZE
1	(6)	CU#10AWG -	PV WIRE/USE-2	N/A	N/A
	(1)	CU #6AWG -	BARE COPPER IN FREE AIR		
2	(6)	CU#10AWG -	THWN-2 +VE & -VE	IMC OR RMC ONLY ON EXTERIOR, EMT IN ATTIC	3/4"
	(1)	CU #10AWG -	CU,THWN-2 GND		
1A	(2)	CU#10AWG -	PV WIRE/USE-2	N/A	N/A
	(1)	CU #6AWG -	BARE COPPER IN FREE AIR		
2A	(2)	CU#10AWG -	THWN-2 +VE & -VE	IMC OR RMC ONLY ON EXTERIOR, EMT IN ATTIC	3/4"
	(1)	CU #10AWG -	CU,THWN-2 GND		
1B	(4)	CU#10AWG -	PV WIRE/USE-2	N/A	N/A
	(1)	CU #6AWG -	BARE COPPER IN FREE AIR		
2B	(4)	CU#10AWG -	THWN-2 +VE & -VE	IMC OR RMC ONLY ON EXTERIOR, EMT IN ATTIC	3/4"
	(1)	CU #10AWG -	CU,THWN-2 GND		
2C	(6)	CU#10AWG -	THWN-2 +VE & -VE	IMC (6" MIN DEPTH BURIAL)	1"
	(1)	CU #10AWG -	CU,THWN-2 GND		
2D	(2)	CU#10AWG -	THWN-2 +VE & -VE	IMC (6" MIN DEPTH BURIAL)	1"
	(1)	CU #10AWG -	CU,THWN-2 GND		
3	(3)	CU#6AWG -	THWN-2 L1,L2,N	IMC ON EXTERIOR	3/4"
	(1)	CU#10AWG -	CU,THWN-2 GND		
3A	(3)	CU#6AWG -	THWN-2 L1,L2,N	IMC (6" MIN DEPTH BURIAL)	3/4"
	(1)	CU#10AWG -	CU,THWN-2 GND		
4	(3)	CU#3/0AWG -	THWN-2 L1,L2,N	IMC ON EXTERIOR	2"
	(1)	CU#4AWG -	CU,THWN-2 GND		
5	(2)	CU#3/0AWG -	THWN-2 L1,L2	IMC ON EXTERIOR	2"
	(1)	CU#3/0AWG -	THWN-2 N		

NOTE : "CONDUIT SIZE IS MINIMUM REQUIRED PER NEC300.17. CONTRACTOR MAY UPSIZE AS NEEDED".

INTERCONNECTION NOTES:
1. INTERCONNECTION SIZING, LIMITATIONS AND COMPLIANCE DETERMINED IN ACCORDANCE WITH [NEC 705.13], AND [NEC 690.59].
2. GROUND FAULT PROTECTION IN ACCORDANCE WITH [NEC 215.9], [NEC 230.95].
3. ALL EQUIPMENT TO BE RATED FOR BACKFEEDING.
4. PV BREAKER TO BE POSITIONED AT THE OPPOSITE END OF THE BUSBAR RELATIVE TO THE MAIN BREAKER.

DISCONNECT NOTES:
1. DISCONNECTING SWITCHES SHALL BE WIRED SUCH THAT WHEN THE SWITCH IS OPENED THE CONDUCTORS REMAINING LIVE ARE CONNECTED TO THE TERMINALS MARKED “LINE SIDE” (TYPICALLY THE UPPER TERMINALS)
2. AC DISCONNECT MUST BE ACCESSIBLE TO QUALIFIED UTILITY PERSONNEL, BE LOCKABLE, AND BE A VISIBLE-BREAK SWITCH
3. DISCONNECT MEANS AND THEIR LOCATION SHALL BE IN ACCORDANCE WITH [NEC 225.31] AND [NEC 225.32].

RACKING NOTE:
1. BOND EVERY OTHER RAIL WITH #6 BARE COPPER

GROUNDING & GENERAL NOTES:
1. **GROUNDING ELECTRODES AND GROUNDING ELECTRODE CONDUCTORS.** ADDITIONAL GROUNDING ELECTRODES SHALL BE PERMITTED TO BE INSTALLED IN ACCORDANCE WITH 250.52 AND 250.54.GROUNDING ELECTRODES SHALL BE PERMITTED TO BE CONNECTED DIRECTLY TO THE PV MODULE FRAME(S) OR SUPPORT STRUCTURE PER [NEC 690.47(B)]
2. PV INVERTER IS UNGROUNDED, TRANSFORMER-LESS TYPE.
3. DC GEC AND AC EGC TO REMAIN UNSPLICED, OR SPLICED TO EXISTING ELECTRODE
4. ANY EXISTING WIRING INVOLVED WITH PV SYSTEM CONNECTION THAT IS FOUND TO BE INADEQUATE PER CODE SHALL BE CORRECTED PRIOR TO FINAL INSPECTION.
5. JUNCTION BOX QUANTITIES, AND PLACEMENT SUBJECT TO CHANGE IN THE FIELD - JUNCTION BOX DEPICTED ON ELECTRICAL DIAGRAM REPRESENT WIRE TYPE TRANSITIONS.
6. AC DISCONNECT NOTED IN EQUIPMENT SCHEDULE OPTIONAL IF OTHER AC DISCONNECTING MEANS IS LOCATED WITHIN 10' OF SERVICE DISCONNECT.
7. RACEWAYS AND CABLES EXPOSED TO SUNLIGHT ON ROOFTOPS SHOULD BE INSTALLED MORE THAN 7/8" ABOVE THE ROOF USING CONDUIT SUPPORTS.

1 ELECTRICAL LINE DIAGRAM
PV-5 SCALE: NTS

TRON SOLAR

TRON SOLAR LLC.
CONTRACTOR ADDRESS:
950 CORPORATE WOODS PARKWAY,
VERNON HILLS, IL 60061
LICENSE #. TGC110144

REVISIONS		
DESCRIPTION	DATE	REV
INITIAL	07/03/2026	A
REVISION	10/12/2026	B

PROJECT NAME & ADDRESS

ELIZABETH L WYER / ETHAN WYER RESIDENCE
203 W STATE ST, FAIRMOUNT, IL 61841
(40.047315, -87.832734)

DRAWN BY
ESR

SHEET NAME
ELECTRICAL LINE DIAGRAM

SHEET SIZE
ANSI B
11" X 17"

SHEET NUMBER
PV-5